

# STANDARD OPERATING PROCEDURE for EOT CRANE BREAKDOWN

## Immediate Actions (Operator Responsibility)

1. **Wear Proper PPE:** All personnel involved must wear appropriate Personal Protective Equipment (PPE), such as safety helmets, safety shoes, full body harness and cotton hand gloves.
2. **Stop Operation:** Immediately stop crane movement and activate the **Emergency Stop** button.
3. **Warn Others:** Alert all personnel in the vicinity of the breakdown and the potential hazard. Barricade the immediate area to prevent unauthorized access and keep everyone out of harm's way.
4. **Secure the Load (If applicable):** If a load is suspended, the operator must remain at the controls (if safe) to prevent the load from dropping. If possible and safe, lower the load to the ground level and chock it securely.
5. **Remove Rigging Gear:** Disconnect all slings, shackles, and other lifting gear from the hook once the load is on the ground.
6. **Report the Breakdown:** The operator must immediately contact their supervisor and the concerned maintenance/electrical department to inform them of the problem, including any unusual sounds, jerks, load sliding, brake not working or visible damage observed before or during the malfunction.

## Maintenance and Recovery Actions (Qualified Personnel Responsibility)

1. **Assess and Diagnose:** Qualified maintenance technicians/engineers, upon arrival, will conduct a systematic assessment and diagnosis to identify the root cause of the breakdown (e.g., electrical malfunction, brake failure, wire rope damage). They should refer to the manufacturer's manuals for guidance.
2. **Isolate and Lock Out/Tag Out:** The maintenance team must ensure the main power source remains locked out and tagged out (LOTO) to prevent accidental energization during repairs.
3. **Repair/Replace Components:** The team will implement the necessary repairs, replacing any damaged or worn components with genuine or approved replacements.

4. **Test and Verify:** After repairs, the crane must undergo thorough testing (e.g., no-load test, rated load test) to ensure proper functionality and safety before being returned to service.
5. **Document and Analyze:** The failure, the repair steps taken, and any contributing factors (Root Cause Analysis) must be fully documented. The preventive maintenance schedule should be updated to address identified weaknesses and prevent recurrence.

## General Safety Measures

- **Do Not Operate Damaged Equipment:** Never attempt to operate a crane that is damaged or has a suspected malfunction.
- **Qualified Personnel Only:** Only qualified and authorized personnel should perform maintenance or repairs on the crane.
- **No Riding:** Do not allow personnel to ride on the load or hook.
- **Clear Communication:** Use standard hand signals or communication devices to coordinate during troubleshooting or recovery operations.